

COLDPOD · REQUIREMENTS

ColdPod requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish; R15 target \$310, status met against CPD-CAL-001 v0.3
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R2 and R7 targets restated for the liner cut-out and buck-boost driver; status from CPD-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	Record Amish's decisions (CPD-DDR-001); R6 redefined to 16 h, R12 relaxed to 5.5 kg, R15 set to \$300; status from CPD-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

ColdPod requirements

These are the requirements for the concept, checked by calculation at TRL 3 in CPD-CAL-001 v0.3. Amish decided the TRL 2 review items on 2026-09-25 (CPD-DDR-001): R6 is redefined from 24 h to 16 h at 43 °C (D5), R12 is relaxed from 5.0 kg to 5.5 kg (D6), R15 follows the new \$300 budget (D10) and the R10 alarm defaults are adopted (D9). He then accepted the TRL 3 recommendations (CPD-DDR-002): R2 now names the two hardware cut-outs in series and R7 names the buck-boost driver that serves the full 10 to 15 V vehicle range; no target was relaxed. On 2026-09-26 Amish approved a budget top-up to \$310 (CPD-DDR-002 N3), and R15 follows it. Against CPD-CAL-001 v0.3, ten requirements are met, three are at risk, two are **not met** by small margins (R8 and R12) and two cannot be verified at TRL 3 (R10 and R16). Before DDR-002 the count was seven met, five at risk and three not met (R2, R8, R12).

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (CPD-CAL-001)
R1	Keep the payload in the safe range	2 to 8 °C at a buffered probe anywhere in the payload space; control setpoint 5 °C	Heat-transfer calculation; later lab test in a climate chamber	Met on paper: liner held at 2 to 5 °C; the lid pack stays frozen near 2.8 °C on its cold plate in powered hold at 43 °C (was at risk)
R2	Never freeze the payload	No payload-contact surface below 1 °C in normal operation or with a stuck-on Peltier driver; two hardware cut-outs in series in the Peltier supply, on the cold block (–5 °C) and on the liner (3 °C), independent of firmware (CPD-DDR-002)	Calculation of the cold path; fault analysis	Met on paper: with a stuck-on driver the liner settles near 2.0 °C after the liner cut-out opens (was not met, about –4 °C)
R3	Carry a useful load	1.0 L or more of usable payload; holds 20 or more insulin pens up to 170 mm long	Massing model	Met: 1.36 L, 24 pens, up to 172 mm
R4	Hold with no power at all	12 h or more at a constant +43 °C, starting with the PCM frozen	Heat-leak and PCM calculation	At risk: 14.0 h; 10.8 h with 30 % more heat leak
R5	Hold off-grid on a full charge	24 h or more at a constant +32 °C on the internal battery, then the PCM	Energy budget	At risk: 28.4 h; 22.3 h with 30 % more heat leak
R6	Hold off-grid on a full charge in extreme heat	16 h or more at a constant +43 °C on the internal battery, then the PCM; longer hot trips run from external power under R7 (redefined from 24 h, CPD-DDR-001 D5)	Energy budget	At risk: 17.5 h; 13.2 h with 30 % more heat leak
R7	Hold indefinitely on external power	Keeps 2 to 8 °C up to +43 °C from 12 V DC (10 to 15 V, through a buck-boost Peltier driver, CPD-DDR-002) or USB-C PD (20 V, 45 W or more)	Peltier and heat sink calculation	Met on paper: 18.5 W input, well inside 45 W; the buck-boost stage covers a 10 V input; the lid pack stays frozen (was at risk)
R8	Recharge the cold quickly	Refreeze a fully melted PCM in 8 h or less at 25 °C ambient, box empty, while charging the battery	PCM and Peltier calculation	Not met: 8.4 h for all the PCM (jacket 6.9 h, lid pack 8.4 h). The lid pack now has a cold path, so it counts; before, the jacket alone took 5.5 h and the lid pack could not be refrozen
R9	Record the temperature	Payload probe accuracy ± 0.5 °C; log every 1 min; 60 days or more on board; export as CSV over Bluetooth Low Energy or USB	Datasheet and storage calculation; later calibration	Met: 1.38 MB of 2.10 MB; accuracy by sensor selection
R10	Raise alarms early	Local sound, light and display alarm, plus a phone notification when paired: warn after 10 min outside 2 to 8 °C; alarm at once at 0 °C or lower on the payload probe; early freeze warning at 1 °C on the liner probe; low battery, sensor fault and lid open for more than 2 min (defaults	Design review of firmware sketch	Not verifiable at TRL 3: design intent only, no firmware sketch

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (CPD-CAL-001)
		decided, CPD-DDR-001 D9; adjustable per product)		
R11	Keep logging when the cooling stops	Logger runs 14 days or more after the Peltier is shut off for low battery	Power budget	Met: about 96 days
R12	Be light enough to carry all day	5.5 kg (12.1 lb) or less empty (relaxed from 5.0 kg, CPD-DDR-001 D6)	Mass estimate, later weighing	Not met: 5.53 kg (was 5.60 kg), after thinner printed parts and the lid cold plate
R13	Be compact	Fits in 400 x 250 x 250 mm including handle	Massing model	Met: 366 x 212 x 207 mm
R14	Travel by air	Battery 100 Wh or less, removable or with a shipping switch	Battery specification	Met: 76.8 Wh
R15	Be affordable and buildable	Parts cost \$310 or less (raised from \$300, CPD-DDR-002 N3, decided by Amish, 2026-09-26); no custom PCB for the first build	Priced BOM	Met: \$303 against \$310 (was not met against \$300), after the lid cold plate, liner cut-out and buck-boost driver
R16	Survive field use	Splash resistant (IP54 target for electronics bays); survives a 0.5 m drop onto a hard floor while loaded	Design review; later test	Not verifiable at TRL 3
R17	Be clearly labelled as a prototype	"Research prototype, not a medical device" on the box, the display start screen and every exported log	Design review	Met by design

■ Assumptions

- Ambient cases follow the WHO PQS test point of +43 °C for the hot zone ([WHO PQS E004/VC01](#)), +32 °C for a typical hot day and 25 °C for a clinic.
- The 2 to 8 °C window and the no-freezing rule come from insulin and vaccine storage guidance ([US FDA](#); [CDC Pink Book, chapter 5](#)).
- Logging and alarm targets are stricter than a WHO 30-day recorder (−0.5 °C for 60 min, +8 °C for 10 h; [TechNet-21](#)) because a carrier in transit can move out of range much faster than a refrigerator.
- R6 matches the WHO long-range carrier cold life only loosely: a WHO carrier is tested passive for 30 h at 43 °C. ColdPod does not claim WHO equivalence.
- The alarm thresholds in R10 were decided by Amish on 2026-09-25 (CPD-DDR-001 D9) and will be adjustable per product.
- Hold-time status uses a ±30 % band on the heat leak: met if the target holds at +30 %, at risk if it holds only at the nominal estimate (CPD-CAL-001).
- R15 counts ColdPod parts only; the payload is not costed.